SUPPLEMENTARY MATERIALS: Groups Are Factors: Learning Linear PDEs by Clustering Modes on the Factors of the Symbol

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These supplementary materials accompany the paper and are not needed to follow it. They contain the notation (section SM1); the proofs of the main results, written to be read rather than only checked (section SM2); the interaction of terms inside a factor (section SM3); the complete algorithm and its parameters (section SM4); further cases and a guide to reading the output (section SM5); what carries over beyond linear equations with constant coefficients (section SM6); and the contents of the code and data archive (section SM7). Numbers without the prefix SM refer to the paper.

SM1. Notation.

SM2. Proofs of the main results. We prove in full the results the algorithm rests on: the factor decomposition (Theorem 2.3), the stability of the split (Theorem 2.6), its Bayesian form (Proposition 3.2) and the counting condition (Proposition 3.1). For identification and finite data (Theorem 2.8 and Proposition 2.9) we give the idea and an example, and prove the part that matters for the algorithm. Complete proofs of every statement, in a longer form, are in the analytical notes of the repository [SM6].

Throughout, $\Omega \subset \mathbb{R}^2$ is open and connected, $\mathcal{A}$ is $C^\infty(\Omega)$ or the distributions $\mathcal{D}'(\Omega)$, and a polynomial $Q \in \mathcal{R}$ acts on $\mathcal{A}$ by $\tau \mapsto \partial_t$, $\xi \mapsto \partial_x$. For a point $z = (\tau_z, \xi_z) \in \mathbb{C}^2$ we write $\psi_z(t,x) = e^{\tau_z t + \xi_z x}$.

SM2.1. Four facts from algebra. The proofs use four classical facts about polynomials in two variables [SM2, SM4]. We state them in the form used and say why each is true.

(F1) *Unique factorization.* Every polynomial is a product of irreducible ones, uniquely up to order and constants. Consequently, if Q_1 and Q_2 have no common factor, a polynomial divisible by both is divisible by $Q_1 Q_2$: $(Q_1) \cap (Q_2) = (Q_1 Q_2)$.

(F2) *Coprime curves meet in finitely many points.* If Q_1, Q_2 have no common factor, $V(Q_1) \cap V(Q_2)$ is finite. (Viewed as polynomials in τ with coefficients rational in ξ, they have a Bézout identity $aQ_1 + bQ_2 = r(\xi)$ with a nonzero polynomial r; at a common zero $r(\xi) = 0$, which leaves finitely many ξ, and symmetrically finitely many τ.)

(F3) *Finitely many common zeros means finite codimension.* If an ideal I has a finite zero set, the quotient $\mathcal{R}/I$ is a finite-dimensional vector space (Hilbert's Nullstellensatz). In particular $\dim \mathcal{R}/(Q_1, Q_2) < \infty$ for coprime Q_1, Q_2.

(F4) *Intersection numbers add.* For coprime Q_1, Q_2, $\dim \mathcal{R}/(Q_1, Q_2)$ is the number of intersection points of the two curves counted with multiplicity, and the multiplicities are additive: the intersection of Q_1 with $Q_2 Q_3$ is the intersection with Q_2 plus the intersection with Q_3 [SM4, Ch. 3].

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Table SM1

Notation. Each symbol has one meaning in the paper and in these materials.

Symbol	Meaning
$\mathcal{R} = \mathbb{C}[\tau, \xi]$	constant-coefficient operators; $\tau \leftrightarrow \partial_t$, $\xi \leftrightarrow \partial_x$
$P,\ Q_j,\ P_i,\ e_i$	an operator and its symbol; a factor; an irreducible factor and its multiplicity
$V(P)$	characteristic variety, the zero set of P in $\mathbb{C}^2$
$\kappa,\ \lambda(\kappa)$	real wavenumber ($\xi = i\kappa$); temporal rate of the mode $e^{\lambda t + i\kappa x}$
$\Lambda_j(\xi)$	branch of an evolution factor $Q_j = \tau - \Lambda_j(\xi)$
$\mathrm{Ann}(u)$	annihilator: all operators Q with $Q(\partial)u = 0$
EP	exponential-polynomials, finite sums $\sum p_j(t,x)e^{\tau_j t + \xi_j x}$
$N,\ \delta(P)$	number of factors; dimension of the overlap of their solution spaces
$\mathcal{S},\ \gamma_N$	sum operator $(u_1, \dots, u_N) \mapsto \sum_j u_j$; its conditioning constant
$c_F,\ \theta_F$	Friedrichs cosine and angle; for two factors $\gamma_2 = 1 - c_F$
$E_{\mathrm{int}},\ \rho$	interaction term; interaction ratio $\|E_{\mathrm{int}}\|/\|u(T) - u_0\|$
w_j	energy share of group j
$\sigma_n,\ \sigma_\kappa$	noise standard deviation; its value in one Fourier bin
$\mathcal{K},\ z$	band of usable wavenumbers; detection threshold in units of σ_κ
$A_p,\ \sigma_{\lambda,p}$	amplitude of mode p; standard deviation of its rate
$N_{\mathrm{s}},\ u^{(q)},\ \pi_q$	number of posterior field samples; sample q; its weight
M; $C(m,d)$	a structure (number and degrees of the factors); a class of structures
B_j; ω^2	orthonormal basis of the sampled kernel of Q_j; prior variance of the component coefficients
W; α; k	cloud of factor vectors; Dirichlet-process concentration; number of groups
FIT-COMPATIBLE, ID-KER, MISFIT, UNDECIDED	output labels

SM2.2. Exponential-polynomials. One more fact is about functions: an exponential-polynomial is exactly a function whose derivatives span a finite-dimensional space.

Lemma SM2.1 (Finite-dimension criterion). *A function or distribution w on a connected Ω is an exponential-polynomial if and only if the span R_w of all its derivatives $\partial_t^j \partial_x^k w$ is finite-dimensional. Its exponents are then the joint eigenvalues of ∂_t and ∂_x on R_w.*

Idea of the proof. If $w = p(t,x)e^{at+bx}$, then $\partial_t w = (p_t + ap)e^{at+bx}$ and $\partial_x w = (p_x + bp)e^{at+bx}$: differentiation never raises the degree of p, so all derivatives stay in the finite-dimensional

space of (polynomials of degree at most $\deg p$) $\times e^{at+bx}$. Conversely, if R_w is finite-dimensional, ∂_t and ∂_x act on it as two commuting matrices; their minimal polynomials m_t, m_x give two ordinary differential equations $m_t(\partial_t)w = 0$ and $m_x(\partial_x)w = 0$, whose common solutions on a rectangle are exponential-polynomials, and connectedness extends this to Ω. ∎

The matrix pencil of subsection 3.1 is exactly this eigenvalue computation for ∂_t, one wavenumber at a time.

SM2.3. The factor decomposition.

What has to be shown. For two coprime factors Q_1, Q_2 consider the sum map

$$\Sigma : \ \ker Q_1 \oplus \ker Q_2 \to \ker(Q_1 Q_2), \qquad (u_1, u_2) \mapsto u_1 + u_2.$$

It is well defined, because $Q_1Q_2(u_1 + u_2) = Q_2(Q_1u_1) + Q_1(Q_2u_2) = 0$. Theorem 2.3 for two factors says two things: Σ is *onto* (every solution splits), and its kernel has dimension $\dim \mathcal{R}/(Q_1, Q_2)$ (the ambiguity is the number of intersection points). The kernel is easy: $\Sigma(u_1, u_2) = 0$ means $u_2 = -u_1$ with u_1 killed by both factors, so

$$\ker \Sigma = \{(v, -v) : \ Q_1 v = Q_2 v = 0\}.$$

The common solutions of Q_1 and Q_2 are killed by the whole ideal (Q_1, Q_2), whose zero set is finite (F2), so by Lemma SM2.1 they are exponential-polynomials, one exponential for each common zero (times polynomials at a zero of higher multiplicity). The hard part is that Σ is onto.

The one-variable case, and what goes wrong in two. In one variable two coprime polynomials satisfy Bézout's identity $aQ_1 + bQ_2 = 1$, and then $u = b(\partial)Q_2(\partial)u + a(\partial)Q_1(\partial)u$ is the split: the first piece is killed by Q_1 because $Q_1 bQ_2 u = b\, Q_1Q_2 u = 0$, the second by Q_2. In two variables coprime polynomials need not generate 1: $\tau + c\xi$ and $\tau - \nu\xi^2$ both vanish at $(0, 0)$, so every combination of them vanishes there too. The quotient $\mathcal{R}/(Q_1, Q_2)$, finite-dimensional by (F3), measures exactly how far we are from Bézout. The proof shows that this defect costs nothing for solvability, only the finite ambiguity, provided the domain has no holes.

Step 1: one point, one potential. Fix a point z and ask: given $a, b \in \mathcal{A}$ with

$$(\partial_t - \tau_z)\, b = (\partial_x - \xi_z)\, a, \tag{SM2.1}$$

is there a $v \in \mathcal{A}$ with $(\partial_t - \tau_z)v = a$ and $(\partial_x - \xi_z)v = b$? Condition (SM2.1) is clearly necessary, since the two operators commute. Write $a = \psi_z \tilde{a}$, $b = \psi_z \tilde{b}$, $v = \psi_z \tilde{v}$; because $(\partial_t - \tau_z)(\psi_z \tilde{v}) = \psi_z\, \partial_t \tilde{v}$, the question becomes: given a vector field $(\tilde{a}, \tilde{b})$ with $\partial_t \tilde{b} = \partial_x \tilde{a}$ (no curl), is it a gradient, $(\tilde{a}, \tilde{b}) = (\partial_t \tilde{v}, \partial_x \tilde{v})$? On a simply connected domain the answer is yes, for smooth fields by the Poincaré lemma and for distributions by de Rham's theorem [SM7]. In the language of homological algebra: $\mathrm{Ext}^1_{\mathcal{R}}(\mathcal{R}/\mathfrak{m}_z, \mathcal{A}) = H^1_{\mathrm{dR}}(\Omega) = 0$, where $\mathfrak{m}_z$ is the ideal of polynomials vanishing at z. The solutions of the homogeneous problem, $a = b = 0$, are the multiples of ψ_z, a space of dimension one.

Step 2: finitely many points. Let M be any $\mathcal{R}$-module of finite dimension over $\mathbb{C}$, for instance $M = \mathcal{R}/(Q_1, Q_2)$. Such a module is built from one-point modules $\mathcal{R}/\mathfrak{m}_z$ by finitely many extensions (a composition series). For each extension $0 \to M' \to M \to M'' \to 0$ the long exact sequence of $\mathrm{Hom}(\cdot, \mathcal{A})$ shows that if the solvability problem of Step 1 has no obstruction

for M' and M'', it has none for M, and that the dimensions of the solution spaces add. By induction on $\dim M$,

$$\text{(SM2.2)} \qquad \operatorname{Ext}^1_{\mathcal{R}}(M,\mathcal{A}) = 0, \qquad \dim \operatorname{Hom}_{\mathcal{R}}(M,\mathcal{A}) = \dim_{\mathbb{C}} M.$$

Here $\operatorname{Hom}_{\mathcal{R}}(\mathcal{R}/I,\mathcal{A})$ is simply the space of solutions of all equations in I: a module homomorphism is determined by the image v of 1, and it is well defined exactly when $Q(\partial)v = 0$ for all $Q \in I$.

Step 3: two factors. By (F1) the sequence

$$0 \to \mathcal{R}/(Q_1Q_2) \xrightarrow{r\mapsto(r,r)} \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \xrightarrow{(r,s)\mapsto r-s} \mathcal{R}/(Q_1,Q_2) \to 0$$

is exact: the first map is injective because $(Q_1) \cap (Q_2) = (Q_1Q_2)$, and the kernel of the second consists of pairs (r,s) with $r - s = xQ_1 + yQ_2$, which are the image of $r - xQ_1 = s + yQ_2$. Applying $\operatorname{Hom}_{\mathcal{R}}(\cdot,\mathcal{A})$ turns each quotient $\mathcal{R}/I$ into the solutions of I and gives the exact sequence

$$0 \to \ker Q_1 \cap \ker Q_2 \xrightarrow{v\mapsto(v,-v)} \ker Q_1 \oplus \ker Q_2 \xrightarrow{\Sigma} \ker(Q_1Q_2) \to \operatorname{Ext}^1_{\mathcal{R}}\left(\mathcal{R}/(Q_1,Q_2),\mathcal{A}\right).$$

The last group vanishes by (SM2.2), because $\mathcal{R}/(Q_1,Q_2)$ is finite-dimensional (F3). So Σ is onto, and its kernel, the common solutions, has dimension $\dim \mathcal{R}/(Q_1,Q_2)$, again by (SM2.2).

Step 4: more factors. Let $P = Q_1 \cdots Q_N$ with pairwise coprime Q_j, and $P' = Q_2 \cdots Q_N$. Then Q_1 and P' are coprime (F1), so by Step 3 $\ker P = \ker Q_1 + \ker P'$, and by induction $\ker P' = \sum_{j\geq 2} \ker Q_j$. This gives the decomposition (2.3). The full sum map $\bigoplus_j \ker Q_j \to \ker P$ is the composition of $\bigoplus_{j\geq 2} \ker Q_j \to \ker P'$ (kernel of dimension $\delta(P')$ by induction) and of $\ker Q_1 \oplus \ker P' \to \ker P$ (kernel of dimension $\dim \mathcal{R}/(Q_1,P')$ by Step 3); both are onto, so the kernel dimensions add, and by (F4) $\dim \mathcal{R}/(Q_1,P') = \sum_{j\geq 2} \dim \mathcal{R}/(Q_1,Q_j)$. This is formula (2.4). □

Example SM2.2 (Running example). For $Q_1 = \tau + c\xi$ and $Q_2 = \tau - \nu\xi^2$, substitute $\tau = -c\xi$ into Q_2: $-c\xi - \nu\xi^2 = -\xi(c + \nu\xi)$, with the two simple roots $\xi = 0$ and $\xi = -c/\nu$. So the curves meet at $(0,0)$ and $(c^2/\nu, -c/\nu)$, each once, and $\delta(P) = \dim \mathcal{R}/(Q_1,Q_2) = 2$. The ambiguous pairs are $(v,-v)$ with v a combination of 1 and $e^{(c^2/\nu)t-(c/\nu)x}$: exactly the functions that can be moved between the transport and the diffusion group.

Why the domain matters. Only Step 1 uses the shape of Ω, and it fails on a domain with a hole: on an annulus the field $(\tilde{a},\tilde{b}) = \nabla \arg z$ has no curl but is not a gradient. This is the obstruction behind the example of the Laplacian in subsection 2.2.

SM2.4. Stability of the split.

Proof of Theorem 2.6. The sum operator $\mathcal{S} : \bigoplus_j M_j \to \ker_{H^s} P$ is bounded, by the triangle and Cauchy–Schwarz inequalities, $\|\sum_j u_j\|^2 \leq N \sum_j \|u_j\|^2$. Every u in its range has components in H^s by definition, so "every $u \in \ker_{H^s} P$ has components in H^s" means that the range is all of $\ker_{H^s} P$, i.e. that it is closed and dense. By the closed range theorem [SM1, Thm. 2.21], the range is closed if and only if $\mathcal{S}$ is bounded below on the orthogonal complement of its kernel, which is the statement $\gamma_N > 0$ by the definition (2.6). In that case take, for a given u,

the preimage $v = (u_1, \dots, u_N)$ orthogonal to $\ker \mathcal{S}$, which is the one of minimum norm; then $\|u\|^2 = \|\mathcal{S}v\|^2 \geq \gamma_N \|v\|^2$, which is (2.7). In finite dimension every linear map has closed range, and the range is the span of the sampled exponentials of P, each of which lies in the kernel of one factor. ∎

The two-factor case, in a picture. Why is $\gamma_2 = 1 - c_F$? Take unit vectors $u_1 \in M_1$ and $u_2 \in M_2$. Then $\|u_1 - u_2\|^2 = 2 - 2\,\mathrm{Re}\langle u_1, u_2\rangle$, while the pair has norm $\sqrt{2}$, so the ratio in (2.6) is $1 - \mathrm{Re}\langle u_1, u_2\rangle$. It is smallest when u_1 and u_2 are as aligned as possible, i.e. when $\mathrm{Re}\langle u_1, u_2\rangle = \cos\theta_F$, after excluding the directions the two spaces share, which form $\ker \mathcal{S}$. Subsection SM2.5 makes this exact through the principal angles; for infinite-dimensional spaces see [SM3, Ch. 9].

SM2.5. The angle sets the error bars.

Proof of Proposition 3.2. With orthonormal bases B_1, B_2 of the two sampled kernels,

$$\mathcal{S}^H \mathcal{S} = \begin{pmatrix} I & B_1^H B_2 \\ B_2^H B_1 & I \end{pmatrix}.$$

Take the singular value decomposition $B_1^H B_2 = U\ \mathrm{diag}(\cos\theta_k)\, V^H$; the singular values are the cosines of the principal angles θ_k between the two kernels. In the basis given by the columns of U and V the matrix splits into 2×2 blocks $\left(\begin{smallmatrix} 1 & \cos\theta_k \\ \cos\theta_k & 1 \end{smallmatrix}\right)$, with eigenvalues $1 \pm \cos\theta_k$, plus the eigenvalue 1 on directions of the larger kernel without a partner. The posterior covariance $\Gamma_a = \sigma_n^2(\mathcal{S}^H\mathcal{S} + (\sigma_n^2/\omega^2)I)^{-1}$ is diagonal in the same basis, with variance $\sigma_n^2/(\mu + \sigma_n^2/\omega^2)$ along an eigenvector of eigenvalue μ. The eigenvalue 0 occurs exactly for $\theta_k = 0$, the directions shared by the two kernels, i.e. $\ker \mathcal{S}$; the smallest nonzero eigenvalue is $1 - \cos\theta_{\min} = 1 - c_F$, which gives (3.8). On a shared direction $\mu = 0$, the variance is ω^2, the prior; the likelihood depends only on the sum $a_1 + a_2$ there, so the posterior of the difference is the prior, centred at zero, and the posterior mean splits the shared part equally. ∎

Since $1 - \cos\theta = 2\sin^2(\theta/2)$, the posterior standard deviation in the worst direction is about $\sigma_n/(\sqrt{2}\sin(\theta_F/2))$ while it is above the prior floor: halving a small angle doubles the error bars.

SM2.6. The counting condition.

Proof of Proposition 3.1. Write $\Lambda(\xi) = \sum_{k=0}^{d} \beta_k \xi^k$. The rates at the nodes $i\kappa_\ell$ are $\lambda_\ell = \sum_k \beta_k (i\kappa_\ell)^k$, a linear system $V\beta = \lambda$ with the Vandermonde matrix $V_{\ell k} = (i\kappa_\ell)^k$. With $d+1$ distinct nodes V is square and invertible, so the coefficients are determined, and $\|\delta\beta\| \leq \|V^{-1}\|\,\|\delta\lambda\|$. With only d nodes, the polynomial $\prod_\ell (\xi - i\kappa_\ell)$ of degree d vanishes at all of them and can be added to Λ without changing any observed rate. When the nodes lie within a spread h_κ, the Lagrange basis polynomials that make up V^{-1} have coefficients of size h_κ^{-d} [SM5]. For real coefficients $\mathrm{Re}\,\Lambda(i\kappa) = \sum_k (-1)^k \beta_{2k} \kappa^{2k}$ and $\mathrm{Im}\,\Lambda(i\kappa) = \sum_k (-1)^k \beta_{2k+1} \kappa^{2k+1}$ separate the even and the odd coefficients into two polynomials in κ^2, with $\lfloor d/2 \rfloor + 1$ and $\lceil d/2 \rceil$ unknowns, which is the real count. ∎

SM2.7. Identification and finite data: the ideas.

Identification (Theorem 2.8). The theorem asks when a solution u of $P(\partial)u = 0$ satisfies no equation except the multiples of P. The idea is to remove one irreducible factor P_i and look at what is left, $v_i = (P/P_i)(\partial)u$, which solves $P_i(\partial)v_i = 0$.

- If v_i is an exponential-polynomial, it sits on finitely many points of the curve $V(P_i)$, and some polynomial h that is *not* a multiple of P_i also vanishes at those points and kills v_i. Then $h \cdot P/P_i$ kills u but is not a multiple of P: u satisfies an equation that P does not explain.
- If no v_i is an exponential-polynomial, each factor is "filled" by u along a whole curve, and only multiples of P can vanish on all of these curves.

Example. For the heat operator $P = \tau - \nu\xi^2$ there is only one factor and $v = u$. A single mode $u = e^{-\nu\kappa^2 t + i\kappa x}$ is an exponential-polynomial, and indeed it also solves the pure decay $(\partial_t + \nu\kappa^2)u = 0$, which is not a multiple of the heat operator: one mode cannot tell the heat equation from any other equation whose curve passes through the point $(-\nu\kappa^2, i\kappa)$. The heat kernel $t^{-1/2}e^{-x^2/(4\nu t)}$ is not an exponential-polynomial (its x-derivatives are polynomials of growing degree times the same Gaussian), and every constant-coefficient equation it satisfies is a multiple of $\partial_t - \nu\partial_x^2$. The complete proof follows the two bullets, using (F2), (F3) and Lemma SM2.1.

Finite data (Proposition 2.9). Part (ii) is the one that shapes the algorithm, and its proof is two lines.

Proof of Proposition 2.9(ii). The Taylor polynomial of order M at z_0 has the same derivatives up to order M as u, and polynomial interpolation in two variables matches any finite set of sample values. Both are polynomials, hence exponential-polynomials. By Theorem 2.8 an exponential-polynomial never has $\mathrm{Ann} = (P)$ when P has an irreducible factor, because removing that factor leaves an exponential-polynomial again. So the finite data are equally consistent with a function that does not determine P. ∎

Part (i) is the positive counterpart: if the derivatives of u span a space of dimension at most n, then u is an exponential-polynomial with at most n terms (counted with polynomial degree), and its derivatives of order up to $2n-1$ at one point determine its annihilator; the reason is the same as for a linear recurrence of order n, which is fixed by $2n$ consecutive values. Data never supply such a bound, which is why the algorithm reports compatibility, not identification.

SM3. The interaction of terms inside a factor. A factor can have several additive terms: $\tau + c\xi - \nu\xi^2$ (advection–diffusion) has two. Proposition 2.5 says that such a factor cannot be split into one solution per term; this section measures what is left over.

Consider an evolution $u_t = \mathcal{A}u$ with $\mathcal{A} = \mathcal{A}_1 + \cdots + \mathcal{A}_K$ (the terms), its flow $\Phi(t) = e^{t\mathcal{A}}$ and the flows $\Phi_n(t) = e^{t\mathcal{A}_n}$ of the single terms. If term n acted alone, it would change the initial datum by $\Delta_n u_0 = (\Phi_n(t) - \mathrm{Id})u_0$. The *interaction term* is what the true evolution does beyond the sum of these single-term changes:

$$E_{\mathrm{int}}(t) = u(t) - u_0 - \sum_{n=1}^{K} (\Phi_n(t)u_0 - u_0). \tag{SM3.1}$$

Proposition SM3.1 (Interaction term). *If the terms commute (in particular for constant*

coefficients),

$$E_{\mathrm{int}}(t) = \Big[\prod_{n=1}^{K}(\mathrm{Id}+\Delta_n) - \mathrm{Id} - \sum_{n=1}^{K}\Delta_n\Big]u_0 = \sum_{|E|\geq 2}\Big(\prod_{n\in E}\Delta_n\Big)u_0,$$

the sum running over the subsets $E \subseteq \{1,\dots,K\}$ with at least two elements. For two terms, $E_{\mathrm{int}} = \Delta_1\Delta_2 u_0$.

Proof. Commuting terms give $\Phi = \prod_n \Phi_n = \prod_n(\mathrm{Id}+\Delta_n)$. Expanding the product over subsets and removing the empty set and the singletons gives the formula. ∎

On a Fourier mode every term is multiplication by its symbol $\hat{\mathcal{A}}_n(\kappa)$, so

$$\widehat{E_{\mathrm{int}}}(t,\kappa) = \Big[\prod_n e^{t\hat{\mathcal{A}}_n(\kappa)} - 1 - \sum_n \big(e^{t\hat{\mathcal{A}}_n(\kappa)} - 1\big)\Big]\hat{u}_0(\kappa),$$

which costs $O(K)$ operations per wavenumber, although writing out all cross terms would need $2^K - K - 1$ of them.

Example (advection–diffusion). With the terms $-c\partial_x$ and $\nu\partial_x^2$, $|\widehat{E_{\mathrm{int}}}|/|\hat{u}_0| = |(e^{-ic\kappa t} - 1)(e^{-\nu\kappa^2 t} - 1)| \leq 2$: the interaction is as large as the solution, not a small correction. On the benchmark input the pipeline reports the interaction ratio $\rho = \|E_{\mathrm{int}}(T)\|/\|u(T) - u_0\| = 0.79$, with 90% interval $[0.788, 0.793]$: most of the evolution is interaction.

The interaction cannot be removed stably. On a bounded, non-periodic window the map $u \mapsto E_{\mathrm{int}}$ is not continuous in any Sobolev norm. The idea is to test it on the growing exponentials $u_\zeta = e^{\lambda_\zeta t + \zeta x}$ with real $\zeta > c/\nu$ and $\lambda_\zeta = \nu\zeta^2 - c\zeta$, legitimate solutions on a bounded interval: diffusion alone amplifies them by $e^{\nu\zeta^2 t}$, the true solution grows only by $e^{\lambda_\zeta t}$, and the interaction must cancel the difference, which is larger by the factor $e^{c\zeta t}$, faster than any power of ζ. A model that writes the data as a sum of single-term pieces, with any penalty on the pieces, therefore leaves a remainder on generic data, and that remainder is the interaction. This is why the algorithm reports ρ as a distribution over posterior samples and never uses it to accept or reject a factor.

SM4. The complete algorithm and its parameters. Algorithm SM4.1 lists all steps; Table SM2 lists the parameters a user may need. All other settings are internal and fixed, and none was changed for the large benchmark of subsection 4.2.

SM5. Further cases and how to read the output.

Four more equations. The protocol of subsection 4.1 (grid, 5% noise, bootstrap intervals) on four more equations of the development benchmark gives Table SM3. All four receive the label FIT-COMPATIBLE. The beam is a product of two complex-conjugate factors that are close to each other, so its components carry the largest error amplification of all cases; it is found first as one factor of degree two in τ, $\tau^2 + 0.0903\,\xi^4$, and the second analysis of its component splits it along its two root branches, which together have the same kernel as the parent. Klein–Gordon and telegraph are irreducible: the method returns one group and does not split them, avoiding the spurious splits produced by fitting one polynomial and factoring it afterwards.

Algorithm SM4.1 Learning a constant-coefficient PDE by grouping its modes

Require: observations $\{(t_i, x_i, y_i)\}$ or a noisy grid; noise model; starting class $(m, d) = (3, 4)$

draw N_s posterior field samples $u^{(q)}$ with weights π_q ▷ subsection 3.1
repeat ▷ choice of class, subsection 3.3
  for each sample $u^{(q)}$ **do**
    FFT in x; keep the band $\mathcal{K}$ above $z\,\sigma_\kappa$ and below both Nyquist limits
    matrix pencil per $\kappa \in \mathcal{K}$: modes $(\kappa, \lambda_p, A_p, \sigma_{\lambda,p})$
    for each structure $\mathsf{M} \in C(m, d)$ **do**
      grouping I: assign modes to factors and refit the factors ▷ subsection 3.2
    end for
    $P(\mathsf{M}) \propto e^{-\mathrm{BIC}/2}$; components u_j, shares w_j, γ_N, labels
    analyse each component again (depth ≤ 2)
  end for
  pool over samples; if most mass fails the tests, propose a larger class
until the larger class does not predict held-out modes better, or $(m, d) = (4, 8)$
refit every sample at the consensus structure
grouping II: cloud W of factors; Dirichlet-process clustering ▷ subsection 3.4
posterior of the components, $P(\text{split})$, intervals of w_j
evidence on the raw observations: drop, prune, merge or split factors; goodness of fit ▷ subsection 3.5
return factors with intervals, $P(k)$, $P(\mathsf{M})$, w_j, γ_N, c_F, distribution of ρ, label

Table SM2

Parameters a user may need, with their defaults.

Parameter	Default	Meaning
starting class (m, d)	$(3, 4)$	total degree in τ and degree in ξ of the first class of structures
largest class	$(4, 8)$	largest class the loop may grow to
concentration of $P(\mathsf{M})$	0.9	posterior mass on one structure required to stop growing
test level	$\log_{10} p = -3$	level of the goodness-of-fit tests (label MISFIT, growth of the class)
detection threshold z	3	a Fourier bin is used only above $z\,\sigma_\kappa$
misfit cap	3	a mode farther than 3 standard deviations from all varieties costs a fixed 9
evidence margin	1 nat	needed to drop, merge or split a factor
posterior samples N_s	8	field samples passed to the core
bootstrap refits	16	refits for coefficient intervals on gridded input
DP concentration grid	$10^{-3}, \ldots, 10$	candidates for α, chosen by the evidence lower bound
prior scale grid	$10^{-6}, \ldots, 10^{2}$	candidates for ω^2, chosen by empirical Bayes

Table SM3
Further single runs. "Error" is the largest relative coefficient error.

Equation	Recovered	Groups	Separation	Error
beam	$(\tau - 0.3004\,i\xi^2)(\tau + 0.3004\,i\xi^2)$	2 (conjugate)	$c_F = 0.939$, $\gamma_N^{-1/2} = 4.05$	0.14%
Klein–Gordon	$\tau^2 - 0.9997\,\xi^2 + 0.9996$	1 (irreducible)	–	0.04%
telegraph	$\tau^2 + 0.8008\,\tau - 0.9990\,\xi^2$	1 (irreducible)	–	0.10%
advection–diffusion	$\tau + 1.0002\,\xi - 0.3004\,\xi^2$	1 (two terms)	$\rho = 0.79$	0.14%

SM5.1. How to read the output. The worked example of subsection 4.1, run with the benchmark protocol (one noisy grid, 16 bootstrap refits), produces a text report and five images; all of them are in the folder `example_run/` of the archive, and the report returns the equation of Table 1. This section says how to read them. With scattered data the same report is computed over posterior samples of the field instead of bootstrap refits.

Three kinds of number. The output mixes three kinds of number, which must not be confused. *Posterior probabilities* ($P(k)$, $P(\mathsf{M})$, $P(\text{split})$) come from model comparison and say which model wins within the models considered; they are not p-values. *Goodness-of-fit p-values*, always printed as $\log_{10} p$, test the null hypothesis that the reported equation holds and the residual is white noise of the given level; a small p withdraws the claim (below 10^{-3} the label becomes MISFIT), while a large p only says that no misfit was found, which is why the label is FIT-COMPATIBLE and never "identified". *Distributions over posterior samples* (of c_F, γ_N, ρ, the shares w_j and every coefficient) are summarized by their 5%, 50% and 95% quantiles; read the width of the interval first.

Directions. A factor Q can carry the mode $e^{\lambda t + i\kappa x}$ exactly when $Q(\lambda, i\kappa) = 0$. The image `directions.png` (Figure SM1) therefore shows, for each factor, $\log_{10}|Q(\lambda, i\kappa)|$ over the plane of wavenumber and rate: dark pixels are the modes the factor can carry, and reading a factor means reading its dark curve. Because λ is complex each factor gets two panels, oscillation ($\operatorname{Im}\lambda$) and damping ($\operatorname{Re}\lambda$). A tilted line in the oscillation panel with a flat line at zero in the damping panel is transport; a flat line in the oscillation panel with a downward parabola in the damping panel is diffusion; curves in both are damped waves. A blurred or broken curve means poorly determined coefficients.

Separation of the factors. The image `operator_space.png` (Figure SM2) compares the reported factors and the factors found in the posterior samples by the largest principal angle between their solution spaces. Two compact blocks far apart mean two well separated factors on which the samples agree; a spread-out cloud, or samples halfway between two factors, means that the samples disagree about the structure; two reported factors close together mean a small γ_N and wide intervals for the components.

Components. The image `modes_and_components.png` shows, for each factor, a few of its kernel modes and the component u_j of the data attributed to it; the components add up to the data up to the residual. If a component looks like noise, or like a copy of another, that factor is not well separated. `directions_uncertainty.png` shows where the posterior samples disagree about each curve, and `distributions.png` the histograms of c_F, γ_N and ρ over the samples.

Five questions to ask of any result.

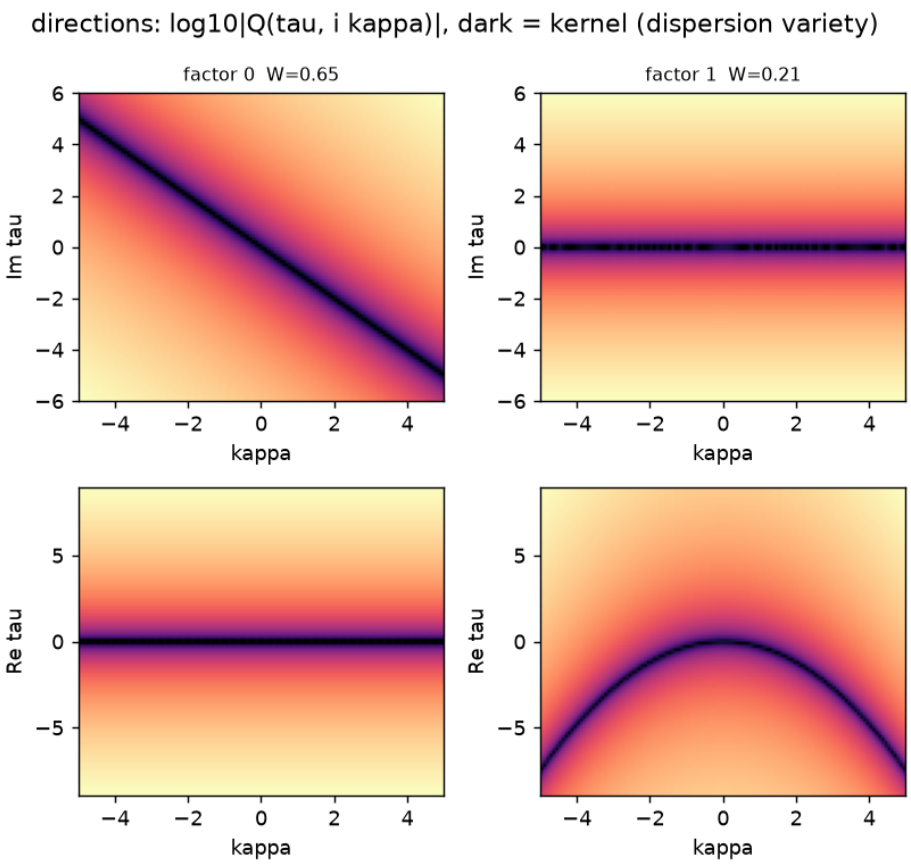


Figure SM1. *`directions.png`: each column is one reported factor; dark curves are the modes it can carry. Left: transport (tilted in the oscillation row, flat in the damping row). Right: diffusion (flat in the oscillation row, a parabola in the damping row).*

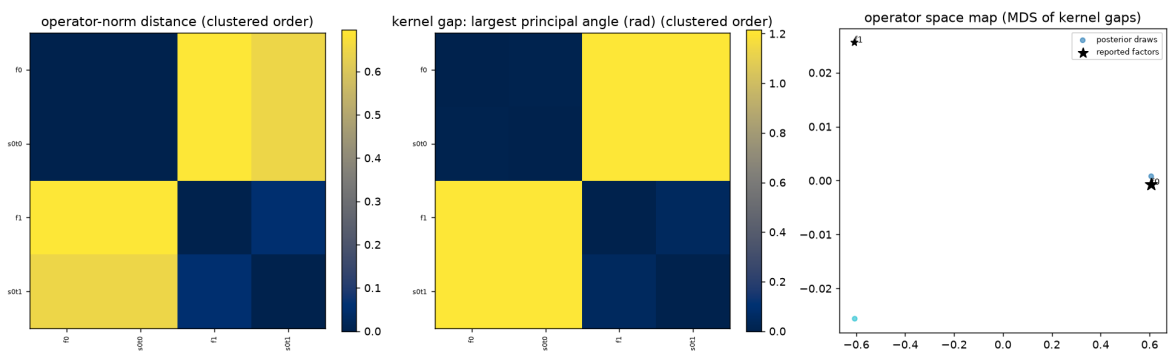


Figure SM2. *`operator_space.png`. Left: distance between operators. Middle: largest principal angle between the solution spaces of every pair of factors (the reported factors `f0`, `f1` and the factors `s0t0`, `s0t1` found in the field sample), ordered by cluster; dark diagonal blocks are the groups, about* 1.2 *rad apart. Right: a two-dimensional map of these angles; stars are the reported factors, dots the factors of the samples.*

1. What is the label? Anything other than FIT-COMPATIBLE: read the note that comes with it.
2. Are the factors separated? $P(\text{split})$ close to one, γ_N not tiny, two distinct blocks in Figure SM2.
3. Is each coefficient narrow and nonzero? Look at the 90% interval of every term.
4. Does the model pass the tests? No $\log_{10} p$ below -3.
5. Do the pictures look like physics? Clean dark curves, components that look like the data, posterior samples on top of the reported factors.

SM6. Beyond linear equations with constant coefficients. The results of the paper rest on one structure: candidate equations form the commutative ring $\mathbb{C}[\tau, \xi]$, solutions form vector spaces, and the exact sequence of subsection SM2.3 turns a factorization of the symbol into a splitting of the solution. Table SM4 summarizes what carries over when this structure is weakened; the paragraphs below explain the three directions in one example each.

Table SM4

What carries over beyond scalar linear equations with constant coefficients.

Setting	Candidate equations	Groups	What survives
linear, constant coefficients	commutative ring $\mathbb{C}[\tau, \xi]$; unique factorization	factors of the symbol, overlap $\delta(P)$	everything in the paper
higher space dimension	$\mathbb{C}[\tau, \xi_1, \ldots, \xi_d]$	hypersurfaces; two factors may share a whole curve	the pencil per wavenumber vector; grouping I
variable coefficients	non-commutative Weyl algebra; no unique factorization	no general splitting theorem	local rates $\lambda(\kappa; x_0)$; identification in a finite class of coefficients
nonlinear	a finite dictionary of jet monomials	none (no superposition)	the interaction term and its short-time expansion

Higher dimension. In d space dimensions the matrix pencil still reads the temporal rates at each wavenumber vector $\kappa \in \mathbb{R}^d$, one small eigenvalue problem per wavenumber, and the groups become hypersurfaces $Q_j(\lambda, i\kappa) = 0$ that grouping I can fit in the same way. What is missing is the analogue of Theorem 2.3: in $d + 1 \geq 3$ variables two coprime factors can share a whole curve of zeros, and the overlap is no longer finite-dimensional.

Variable coefficients. If the coefficients vary slowly in x, a windowed transform gives local rates $\lambda(\kappa; x_0)$, and each group becomes a family of curves indexed by position; grouping I would then fit coefficient functions instead of numbers. The operators now form a non-commutative algebra ($\partial_x\, x - x\, \partial_x = 1$), factorization is not unique, and with arbitrary coefficients one solution identifies nothing: any u without zeros solves $(\partial_x - u_x/u)u = 0$. Identification is possible only inside a fixed finite-dimensional class of coefficients; for $iu_t = -u_{xx} + V(x)u$, for instance, $V = (iu_t + u_{xx})/u$ wherever $u \neq 0$.

Nonlinear equations. For $u_t = \sum_n \mathcal{A}_n(u)$ the definition (SM3.1) of the interaction still makes sense, and for short times

$$E_{\mathrm{int}}(t) = \frac{t^2}{2} \sum_{n \neq m} \mathcal{A}_n'[u_0](\mathcal{A}_m(u_0)) + O(t^3),$$

with $\mathcal{A}_n'$ the Fréchet derivative: the flow is its linearization plus a quadratic interaction. For Burgers' equation $u_t = -uu_x + \nu u_{xx}$ this gives $E_{\mathrm{int}}(t) = -\nu(u_0 u_0''' + 2u_0' u_0'')\, t^2 + O(t^3)$, valid while the solution is smooth. So modes of the linearized flow can still be grouped on short windows, and the size of E_{int} tells whether the groups persist. Without a ring there is no analogue of $\delta(P)$, of the projectors (2.5) or of γ_N; in each direction the first task is therefore the one carried out in the paper for the linear case.

SM7. Code and data. The archive `code_and_data.zip` is a snapshot of the repository [SM6] at the version that produced every number of the paper; later versions are maintained there. Its file `CONTENTS.txt` lists every file. In short:

- `code/pdefactor/`: the complete algorithm as a Python package (entry point `run_loop`); `python -m pdefactor.demo` runs it on a simulated equation in under a minute.
- `example_run/`: the worked example of subsection 4.1, run with the benchmark protocol, with the full text report and the five images described in subsection SM5.1, so the output can be inspected without running anything.
- `benchmarks/`: the scripts of both benchmarks and all their stored per-input results, among them `out/large.jsonl` (one row per random equation, with the true factors and both methods' answers).
- The folder `claude/benchmark_tables/` holds the benchmark tables, among them `large.md` and `p8_final_v5.md`, from which Tables 2 and 3 are taken (the folder keeps its name from the repository so that the scripts run unchanged).
- `code/experiments/`: the experiment scripts and those that produce the figures of the paper from the stored results; `tests/`: the test suite.

The `README.md` of the archive gives the commands to install the package, rerun the example, reproduce both benchmarks and regenerate the figures.

REFERENCES

[1] Haim Brezis. *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Universitext. Springer, New York, 2011.

[2] David A. Cox, John Little, and Donal O'Shea. *Ideals, Varieties, and Algorithms*. Undergraduate Texts in Mathematics. Springer, Cham, 4 edition, 2015.

[3] Frank Deutsch. *Best Approximation in Inner Product Spaces*. CMS Books in Mathematics. Springer, New York, 2001.

[4] William Fulton. *Algebraic Curves: An Introduction to Algebraic Geometry*. 2008. Electronic edition of the 1969 book (W. A. Benjamin, New York).

[5] Walter Gautschi. On inverses of Vandermonde and confluent Vandermonde matrices. *Numerische Mathematik*, 4:117–123, 1962.

[6] Flavio Pricoco. pde-factor-fit: reading the factor structure of linear PDEs from data. https://github.com/FlavioApric/pde-factor-fit, 2026. Software and benchmark data.

[7] Georges de Rham. *Variétés différentiables: formes, courants, formes harmoniques*. Hermann, Paris, 1955.